

Project Report

1. INTRODUCTION

1.1 Project Overview

Product placement is a marketing strategy where brands are naturally featured in movies, television shows, web series, and social media content. Instead of using traditional advertisements, products are integrated into entertainment content to increase brand awareness and influence consumer behavior. This project analyzes how strategic product placement affects customer engagement, brand recognition, and purchase decisions. It also examines different media platforms and evaluates how effective product placement is in influencing consumers.

1.2 Purpose

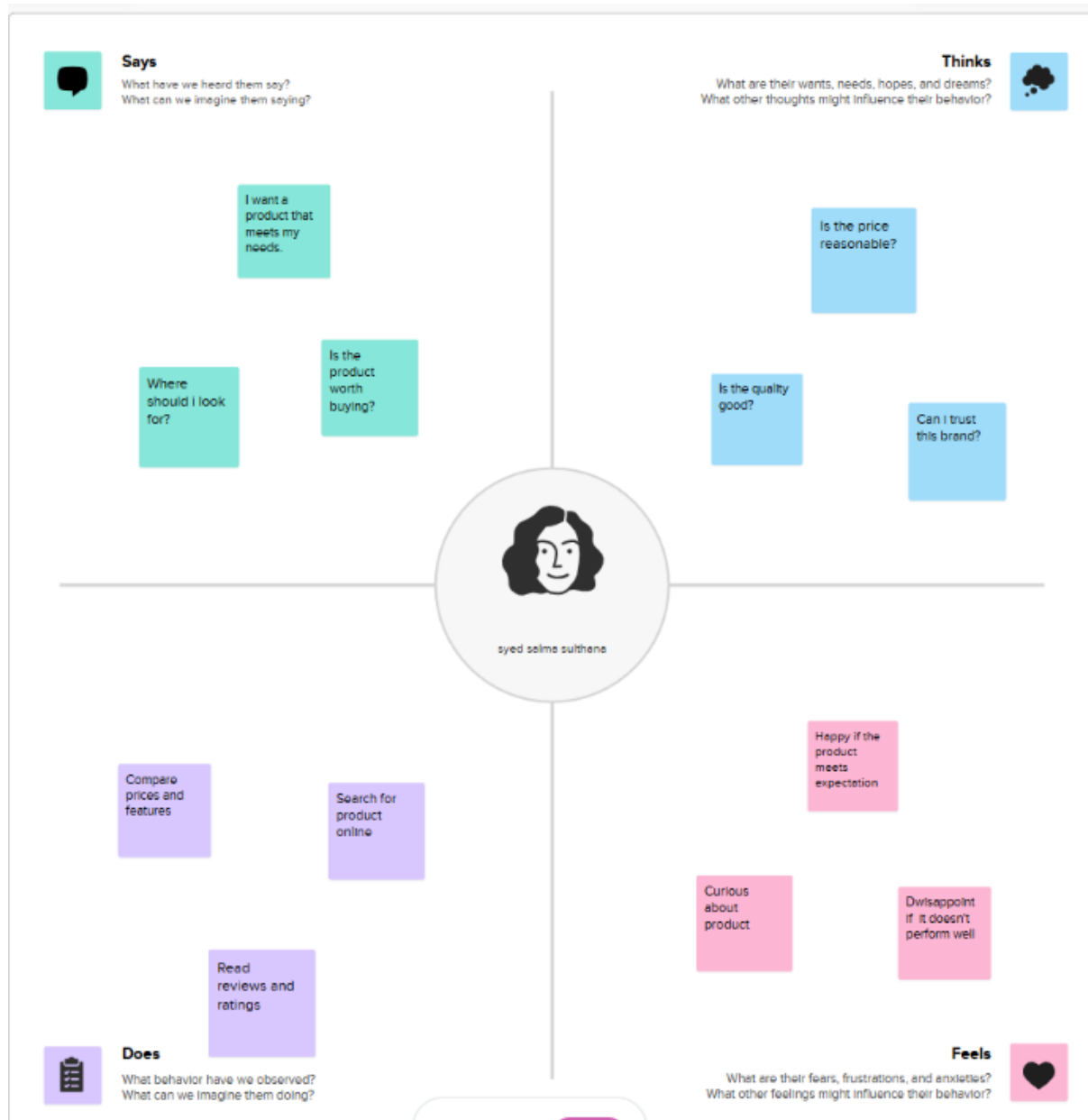
The purpose of this project is to study the impact of product placement on consumer purchasing behavior and brand awareness. The project aims to understand how different forms of media influence customer engagement and buying decisions while identifying the advantages and challenges of product placement as a marketing strategy.

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement Template				
I am	I'm trying to	But	Because	Which makes me feel
I am Business Analyst Sales Manager Marketing Manager Product Manager Data Analyst	I'm trying to Compare competitor pricing Increase product visibility Increase sales and revenue Create interactive dashboard Analyze product sales performance	But Trends are difficult to identify Data contains missing values Large datasets are time-consuming to process Decision-making lacks visual insights Reports are scattered across multiple sources	Because Data is unorganized Manual analysis is slow Data quality issues affect analysis Multiple factors influence sales Traditional reports are difficult to interpret	Which makes me feel Confused Frustrated Uncertain Inefficient Less confident

2.2 Empathy Map Canvas



2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

10 minutes to prepare
1 hour to collaborate
2-8 people recommended

Before you collaborate

Before starting the brainstorming session, each team member reviews the agreed problem statement, explores the product placement options, and identifies key factors affecting product placement, such as pricing, promotions, customer behavior, and competitor influence. Team members share prepared only ideas and suggestions for discussion.

10 minutes

Team gathering

Muskaan Aliyah Shaik
Syed Salma Suthana
Shukh Aysha
Uppara Thanuja
Pravalika Boda

Set the goal

Develop innovative ideas to improve the SkinWaves platform by making skill learning, certification, and career opportunities more accessible, personalized, and engaging for students and professionals.

Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and productive session.

Open article →

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

PROBLEM

How might we identify the best product placement strategy to increase sales, improve customer engagement, and maximize business profit?

Key rules of brainstorming

To run an smooth and productive session

- Stay in topic.
- Defer judgment.
- Go for volume.
- Encourage wild ideas.
- Listen to others.
- If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP You can select a sticky note and list the generated ideas to stick! Don't start drawing!

Person 1

- Place frequently purchased products at eye level.
- Group related products together.
- Display discounts clearly.

Person 2

- Increase sales by optimizing shelf space.
- Reduce slow-moving inventory.
- Arrange products based on demand.

Person 3

- Use attractive promotional displays.
- Bundle complementary products.
- Create eye-catching signage.

Person 4

- Analyze historical sales data.
- Compare sales by shelf position.
- Predict high-performing placements.

Person 5

- Maximize overall profit.
- Increase repeat purchases.
- Make data-driven placement decisions.

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

TIP Add customer behavior to story notes to make easier to find, compare, organize, and categorize. Visual notes to stories within your story.

Group 1: Product Placement

- Shelf Position Analysis
- Eye-Level Product Placement
- Category-Based Arrangement
- Store Layout Optimization
- Product Visibility

Group 2: Customer Behavior

- Customer Buying Patterns
- Purchase Frequency
- Customer Preferences
- Shopping Habits
- Customer Satisfaction

Group 3: Sales & Performance

- Sales Data Analysis
- Best-Selling Products
- Seasonal Sales Trends
- Revenue Growth
- Sales Comparison

Group 4: Marketing Strategy

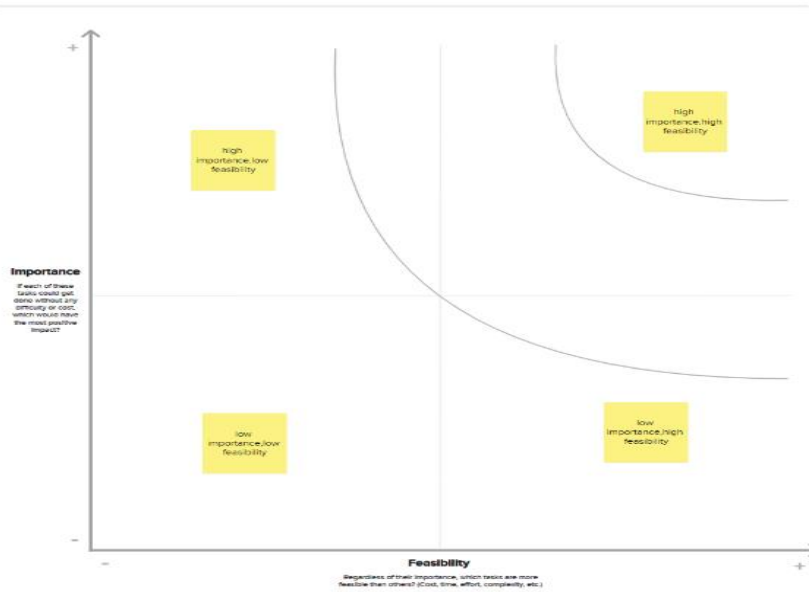
- Promotional Displays
- Discount Offers
- Bundle Products
- Attractive Signage
- Competitor Price Analysis

Group 5: Data Analytics & Decision Making

- Tableau Dashboard
- Sales Reports
- Data Visualization
- Predictive Analysis
- Placement Recommendations

Step-3: Idea Prioritization

TIP Participants can use their cursors to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the laser pointer holding the H key on the keyboard.



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

MY PROJECT >

Strategic Product Placement Analysis



1 STAGE IDEAL Browsing, considering, and evaluating my product payment strategy	2 Enter How do I capture attention and spark interest?	3 Engage Is the core journey, what matters most to me?	4 Exit What do I take away, and how do I feel about the experience?	5 Extend What happens after I leave or stop engaging?
1 Steps What is the customer typically experiencing? Does my project online or offline Refines my changed benefit Reacts to short value statement Feels outcome and value, wants more Learns on my website or demo Compares with alternatives Views key information Has questions before deciding	2 Interactions What touchpoints and channels are used? People who are interested with? • Where, when, how, on what? • Why? What value is material support? Receives quick pitch & asks Refers to & word of mouth Seeks engine results Learning page or website Receives product page Like and / or bookmark Structure, presentation Engage in inquiries	3 Engage Is the core journey, what matters most to me? Views, feels my offering Interacts with my content or team Explores core features or benefits Excludes how I connect from next Processes the information Decides next step Reviews the outcome Drinks or saves for later	4 Exit What do I take away, and how do I feel about the experience? Provides for guides Recommends to others Provides the information Decides next step Reviews the outcome Drinks or saves for later	5 Extend What happens after I leave or stop engaging? Provides for guides Recommends to others Provides the information Decides a review or feedback Discusses a repeat order or supporter Learns a review or feedback
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3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

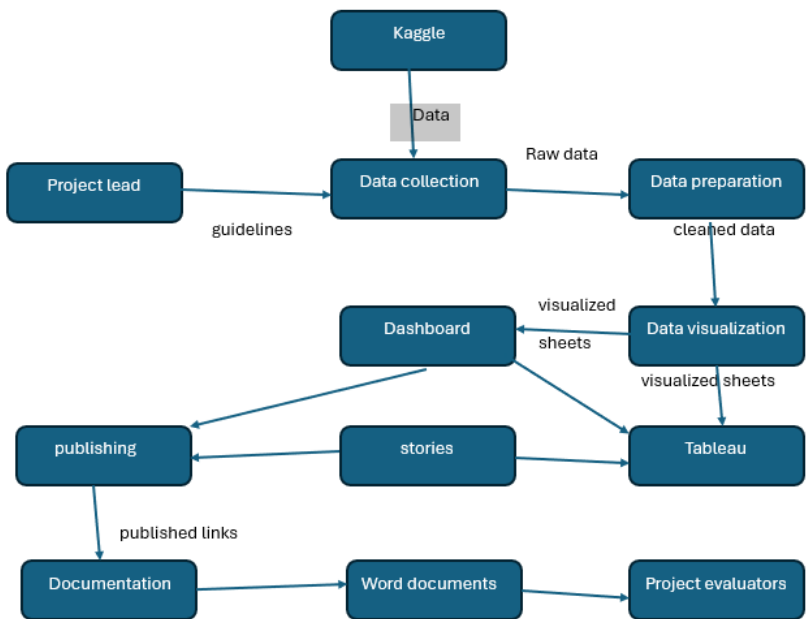
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Collect the product placement dataset from Kaggle. Import the dataset into Tableau.
FR-2	Data Preparation	Clean missing values. Remove duplicate records. Filter and transform the dataset.
FR-3	Data Analysis	Analyse product categories. Compare sales with competitor pricing. Evaluate the impact of promotions on sales.
FR-4	Data Visualization	Create Bar Charts. Create Line Charts. Create Scatter Plots. Create Pie Charts.
FR-5	Dashboard Development	Design an interactive Tableau dashboard. Add filters and KPI indicators. Display key business insights.
FR-6	Report & Story Generation	Create a Tableau Story. Generate reports and summarize strategic product placement insights. Prepare the final dashboard for presentation.

Non-functional Requirements:

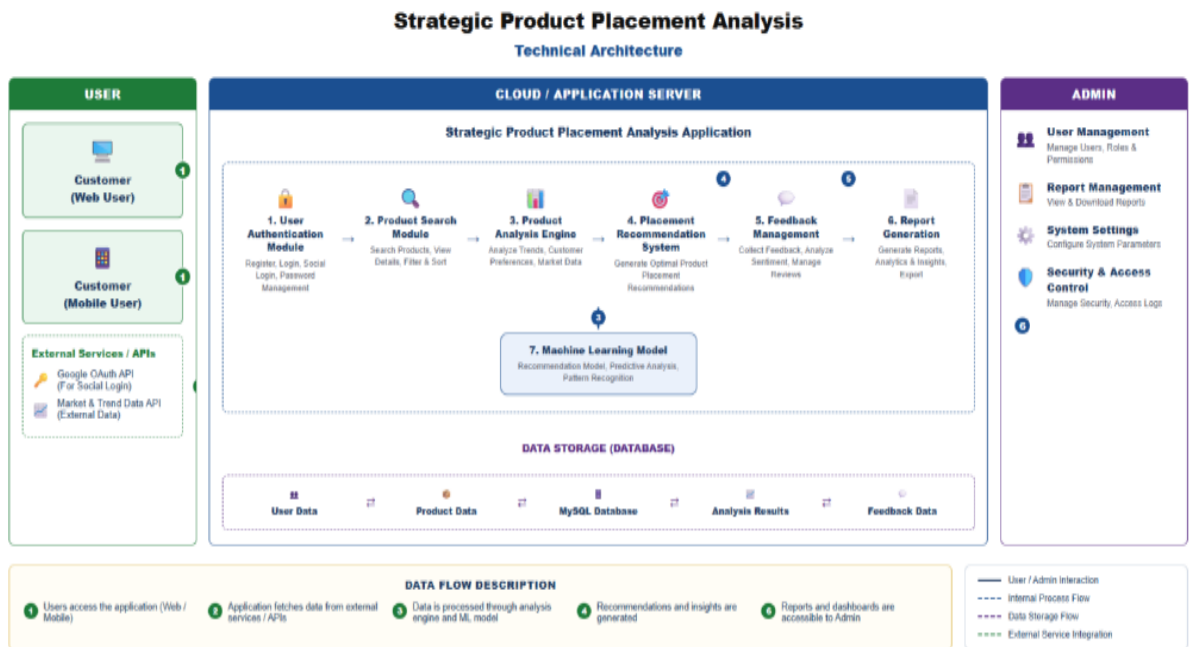
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to understand and navigate for all users.
NFR-2	Security	Project data should be protected from unauthorized access and Modification
NFR-3	Reliability	The dashboard should be protected from unauthorized access and Modification.
NFR-4	Performance	Tableau dashboards should load quickly and respond smoothly to user interaction.
NFR-5	Availability	The dashboard and project files should be available whenever users need them.
NFR-6	Scalability	The system should support larger datasets and additional visualization in the future.

3.3 Data Flow Diagram



3.4 Technology Stack



4. PROJECT DESIGN

4.1 Problem Solution Fit

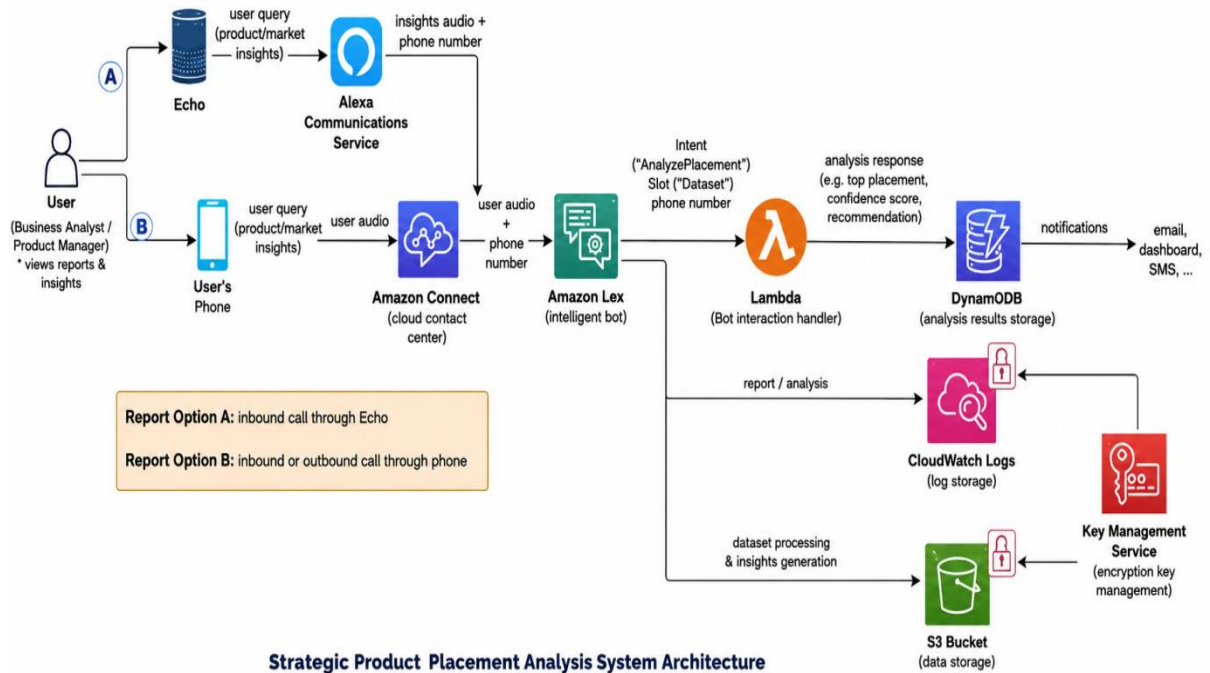
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? - Retail store managers - Merchandising and category managers - Retail business analysts - Supermarket and department store owners - Marketing and sales teams - E-commerce retail businesses	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices? - Limited visibility into the impact of product placement - Lack of data analysis skills and tools - High cost of advanced analytics solutions - Data quality and integration issues - Time constraints to test and optimize placements - Resistance to change from traditional methods	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? - Manual placement based on experience and intuition - Trial-and-error method in stores - Basic sales reports from POS systems - Excel-based analysis - Third-party merchandising agencies - Generic BI dashboards (not placement focused)	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do your customers need to solve? - Identify the best product placement strategy - Understand product performance by location - Increase sales and revenue - Improve customer experience in-store - Optimize shelf space and layout - Make data-driven merchandising decisions	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? - Lack of insights on how placement affects sales - Inability to analyze large volumes of retail data - Multiple factors affect sales (price, promo, location, demand) - No predictive insights for future performance - Lack of tools that combine analytics and recommendations - Changing customer preferences and trends	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? - Relies on past experience and intuition - Uses basic reports and spreadsheets - Follows competitor or market trends - Frequently changes product placement - Asks for feedback from customers or staff - Monitors sales manually after changes	Focus on J&P, lap into BE, understand RC
Identify strong TR EM	3. TRIGGERS TR What triggers customers to act? i.e. reason their pain/gain is activated. - Declining sales or slow-moving products - Low product visibility or customer engagement - Seasonal changes or new product launches - Competitor promotions and offers - Need to improve store layout and efficiency	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, then the new canvas, and check how much it fits here. Strategic Product Placement Analysis System - Collects and integrates sales, product, customer and store data - Analyzes product performance by location and category - Uses machine learning to predict best product placement - Provides actionable recommendations for shelf optimization - Interactive dashboards and visual reports - Helps increase sales, product visibility and customer satisfaction	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE - Web application / dashboard - Email reports and alerts - Website and customer portal - Social media (LinkedIn, Facebook) - Online tutorials and help guides - Webinars and demo sessions 8.2 OFFLINE - In-store meetings and presentations - Workshops and training sessions - Printed reports and manuals - Phone support and consultations - Industry conferences and events	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? Before: Unsure, confused, worried about low sales and wrong decisions After: Confident, satisfied, clear insights, higher sales and happy customers			

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	It is difficult to identify the best product placement to increase sales.
2	Idea / Solution Description	The system analyzes retail sales data and suggests better product placement strategies
3	Novelty / Uniqueness	Uses machine learning and data analysis to improve product placement decisions.
4	Social Impact / Customer Satisfaction	Helps customers find products easily and improves their shopping experience
5	Business Model (Revenue Model)	Subscription plans and premium analysis services.

6	Scalability of the Solution	The system can support more users and businesses in the future.
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4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As an analyst, I can collect retail sales and product placement data from different sources for analysis.	3	High	Shaik Muskaan Aliyah
Sprint-1	Data Preprocessing	USN-2	As an analyst, I can clean the dataset by handling missing values, duplicates,	5	High	Syed Salma Sulthana

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
			and inconsistent records.			
Sprint-1	Exploratory Data Analysis	USN-3	As an analyst, I can analyze sales trends and product performance using visualizations.	5	High	Shaik Muskaan Aliyah
Sprint-1	Feature Engineering	USN-4	As an analyst, I can create new features to improve the accuracy of the product placement analysis.	3	Medium	Shaik Muskaan Aliyah
Sprint-2	Machine Learning Model	USN-5	As an analyst, I can train a machine learning model to predict the best product placement strategy.	8	High	Thanuja uppara
Sprint-2	Model Evaluation	USN-6	As an analyst, I can evaluate the model using performance metrics to ensure accurate predictions	5	High	Shaik Ayesha
Sprint-2	Dashboard & Visualization	USN-7	As a user, I can view interactive dashboards showing sales trends, product performance, and placement recommendations.	5	High	Pravalika Boda
Sprint-2	Business Recommendations	USN-8	As a manager, I can generate reports with product placement recommendations to	3	High	Syed Salma Sulthana

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
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support better decisions.

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	16	4 Days	29 Jun 2026	02 Jul 2026	16	02 Jul 2026
Sprint-2	21	5 Days	03 Jul 2026	07 Jul 2026	21	07 Jul 2026

Velocity:

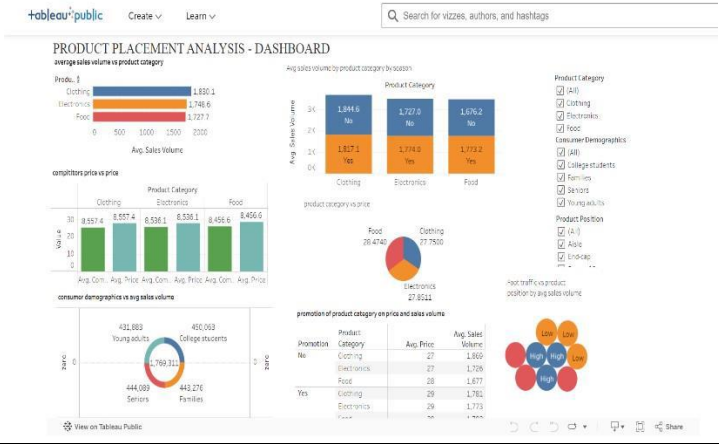
The project was completed in 2 sprints over a period of 9 days. A total of 37 story points were completed, resulting in an average team velocity of 18.5 story points per sprint.

Average Velocity (AV) per day = Total Story Points ÷ Total Duration AV = 37 ÷ 9 = 4.11 story points per day

6. FUNCTIONAL AND PERFORMANCE TESTING

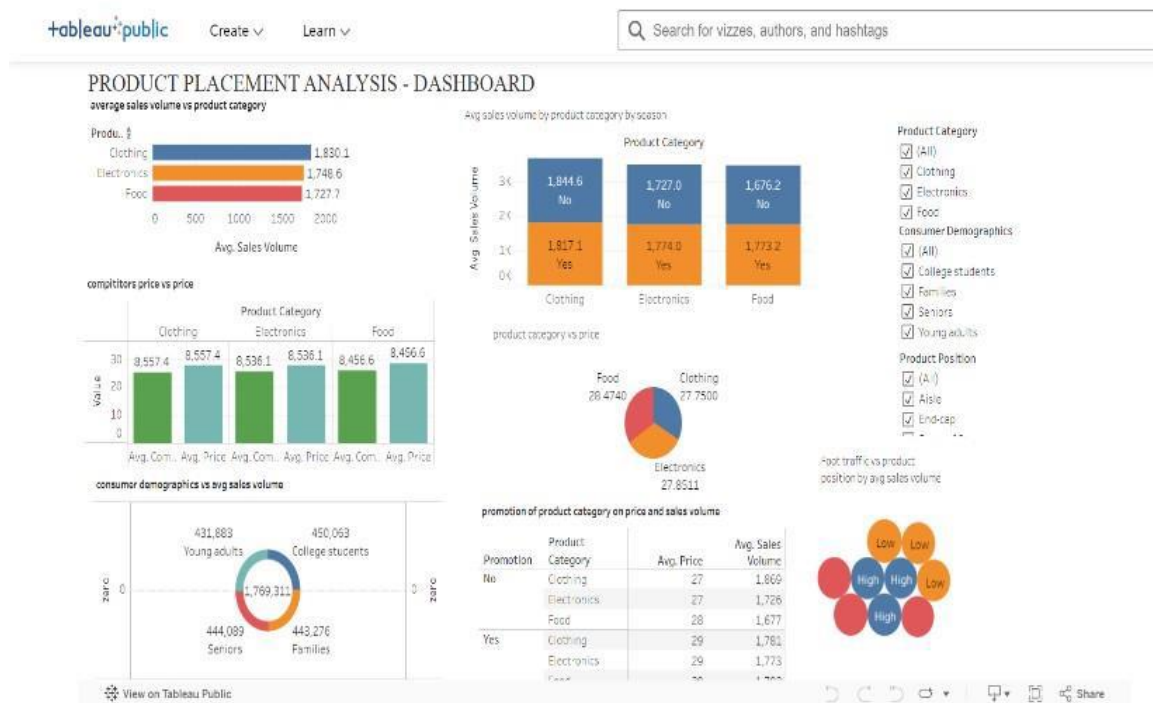
6.1 Performance Testing

S.No	Parameter	Screenshot / Values
1	Data Rendered	Successfully displayed 1,000 records containing 10 attributes. All visualizations, filters, and dashboard elements loaded correctly with complete and accurate data representation.
2	Data Preprocessing	Data preparation included checking data types, identifying and handling missing values, validating numerical and categorical columns, removing duplicate entries where required, and creating

		calculated fields for analysis.
3	Utilization of Filters	Filters Applied: Consumer Demographics, Product Category, and Product Position to provide interactive data exploration and customized insights.
4	Calculated Fields Used	Total Sales Volume, Average Sales Volume, Average Product Price, and Average Competitor Price were created to support performance analysis.
5	Dashboard Design	 The screenshot shows a Tableau dashboard titled "PRODUCT PLACEMENT ANALYSIS - DASHBOARD". It features several visualizations and filters. The top left visualization is a horizontal bar chart titled "average sales volume by product category" showing sales volume for Clothing (1,450.1), Electronics (1,748.6), and Food (1,727.7). Below it is a bar chart titled "competitor price vs price" showing average competitor price and average price for Clothing, Electronics, and Food. The bottom left visualization is a bar chart titled "consumer demographics vs avg sales volume" showing average sales volume for Young adults (421,889), College students (450,093), Seniors (444,093), and Families (440,276). The top right visualization is a bar chart titled "product category vs price" showing average price for Clothing (1,844.8), Electronics (1,727.6), and Food (1,576.2). The bottom right visualization is a pie chart titled "product category vs price" showing the distribution of product categories. The dashboard also includes a table titled "promotion of product category on price and sales volume" and a bubble chart titled "heat map of product position by avg sales volume". The dashboard has a search bar at the top and a list of filters on the right side.
6	Story Design	No of Visualizations / Graphs -8 The story combines all dashboard visualizations to explain how product placement, pricing strategy, promotional activities, consumer demographics, seasonal trends, and foot traffic influence overall sales performance.

7. RESULTS

7.1 Output Screenshots



Dashboard:

https://public.tableau.com/views/productplacementanalysis-dashboard/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story:

https://public.tableau.com/views/productplacementanalysis-story/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

1. ADVANTAGES & DISADVANTAGES

Advantages

- Helps analyze product placement performance across different store locations and product categories.
- Enables data-driven decision-making through interactive dashboards and visual analytics.

- Allows quick identification of high-performing and low-performing products.
- Simplifies complex sales and placement data using easy-to-understand visualizations.
- Assists businesses in optimizing shelf space to improve sales performance and operational efficiency.
- Supports better marketing strategies, inventory management, and demand planning.
- Enhances overall business performance by identifying trends and customer purchasing patterns.
- Reduces manual analysis time by providing automated insights and performance reports.

Disadvantages

- The accuracy of analysis depends heavily on data quality; missing or incorrect data can lead to unreliable results.
- Requires basic knowledge of Tableau to create, interpret, and customize dashboards effectively.
- Limited real-time data analysis when dashboards are not connected to live data sources.
- Provides insights and visualizations but does not automatically recommend business decisions; human interpretation is still required.
- Dashboard performance may decrease when handling very large datasets.
- Results are dependent on the available historical data and may not accurately predict future trends without advanced predictive models.
- Regular data updates are required to ensure reports remain accurate and relevant.

2. CONCLUSION

This project successfully analyzes product placement performance using Tableau dashboards and data visualization techniques. By cleaning, organizing, and analyzing the dataset, valuable insights were obtained regarding product category performance, sales trends, and the effectiveness of product placement strategies. The interactive dashboard enables users to easily monitor key performance indicators (KPIs), compare product performance across different categories and locations, and understand the impact of product placement on sales. Overall, the project demonstrates how data visualization and business intelligence tools can support informed decision-making, optimize product placement strategies, improve inventory planning, and enhance overall retail business performance.

3. FUTURE SCOPE

- > Integrate real-time sales data to enable live dashboard updates and real-time performance monitoring.
- > Apply machine learning algorithms to predict optimal product placement and improve sales forecasting.
- > Expand the analysis to include customer behavior, buying patterns, and customer segmentation.

- > Incorporate advanced analytics techniques, such as product clustering based on sales performance and demand.
- > Develop a mobile-friendly dashboard for easy access across smartphones and tablets.
- > Automate the data cleaning, processing, and reporting workflow to reduce manual effort and improve efficiency.
- > Enhance the dashboard with predictive analytics, alerts, and customizable reports for better decision-making.
- > Integrate data from multiple retail channels (both online and offline) to provide a comprehensive view of product performance.

11. APPENDIX

Source Code (if any)

No source code

Dataset Link:

<https://www.kaggle.com/datasets/amitykulkarni/impact-of-product-positioning-on-sales>

GitHub & Project Demo Link:

Github link:

<https://github.com/233h1a05a1-droid/Strategic-Product-Placement-Analysis>

Demo video link:

<https://docs.google.com/videos/d/1L5kQWI48BNaWNGmrIXca8ke2veCmUo11YfSTAlaT0AU/edit?scene=id.p#scene=id.p>

